Technical Report: Global Sustainability & Energy Tracker

1. Project Summary

This project is an end-to-end Business Intelligence (BI) solution developed to monitor the global shift from conventional energy to green alternatives. By scrutinizing data from **2000 to 2020**, the dashboard identifies the countries leading in sustainability and highlights the correlation between economic growth and environmental impact.

2. The Problem Statement

With India and the rest of the world aiming for "Net Zero" targets, there is a massive need for transparent data. This project solves the problem of "Data Silos" by integrating three critical pillars into a single view:

- **Energy Generation:** Detailed breakup of Renewables, Fossil Fuels, and Nuclear power.
- **Economic Indicators:** Tracking how GDP per Capita influences energy choices.
- **Carbon Footprint:** Measuring the real-world impact through CO2 emissions.

3. Data Engineering (ETL) pipeline

Using the **Power Query M engine**, the raw Kaggle dataset was put through a rigorous cleaning process to ensure the insights are 100% accurate:

- **Handling Nulls & NaNs:** I utilised conditional logic to replace "NaN" strings with 0, ensuring that the mathematical calculations don't break.
- **Data Standardisation:** All energy metrics were converted to a uniform scale of **Terawatt-hours (TWh)** for fair comparison.
- **Geographic Mapping:** Standardised country names to ensure the Global Heatmap accurately represents all regions without errors.

4. Robust Data Modeling

Instead of a messy "flat" table, I implemented a **Star Schema**. This is a professional architecture that ensures the dashboard is fast and scalable:

- **Fact Table:** Fact_Energy_Sustainability (Stores all the core numbers).
- **Dimension Tables:** Dim_Date (Created for time-series analysis) and Dim_Geography (For regional filtering).
- **Relationships:** Established **One-to-Many (1:*)** links using Year and Entity as primary keys.

5. Analytical Logic (DAX)

To move beyond basic charts, I developed custom **Data Analysis Expressions (DAX)** to bring out deeper insights:

- **Sustainability Index:** A custom measure that calculates the percentage of green energy in the total mix.

Green Energy % = ∑ Renewable Energy \ ∑ (Renewable + Fossil + Nuclear)

- **Error Handling:** Used DIVIDE and COALESCE functions to prevent "Divide by Zero" errors—a common issue in data science.
- **Time Intelligence:** Created measures to track Year-over-Year (YoY) growth in renewable capacity.

6. Dashboard Highlights

- **Geospatial Heatmap:** A Red-to-Green gradient map that instantly shows which parts of the world are leadings or lagging.
- **The Transition Trend:** A dual-axis chart that plots the decline of fossil fuels against the rise of renewables over 20 years.
- **Sustainability Leaderboard:** A dynamic Top-10 ranking that automatically updates based on the selected year.

7. Conclusion & Future Scope

The project successfully demonstrates that while global renewable capacity is rising, the transition speed varies significantly by region.